Combined Literature Review: Sign Language Recognition & Fingerspelling Datasets

Merged from both source tables, sorted by year (2026 → 2021), most recent first

S.No	Year	Author(s)	Publisher/Venue	Preprocessing	Techniques & Tools	Result
1	2026	Joyonta Kumar Nath, S.A. Mahadi Alam, Yasir Ullah, Miso Dhar, Joy Dewanjee, Muhammad Umair, Shahid Kamal, Zul Hilmi Abdullah	Scientific Reports (Nature Portfolio)	Images resized to 50x50, normalized to [0,1]; data augmentation (rotation, zoom, translation, horizontal flip) applied only to training set; 80:20 train/validation split	Transfer learning with ConvNeXtXLarge, EfficientNet-B0, VGG19, ResNet-50 (ImageNet pretrained, custom Dense+softmax head); Adam optimizer, categorical cross-entropy, TensorFlow/Keras; t-SNE, Cohen's Kappa, MCC for analysis; MediaPipe Hands for real-time deployment	ConvNeXtXLarge: 99.81% accuracy (best); EfficientNet-B0: 99.68%; VGG19: 99.31%; ResNet-50: 97.29% (on custom 55,500-image ASL alphanumeric dataset, 37 classes)
2	2025	Kirandevraj R, Vinod K. Kurmi, Vinay P. Namboodiri, C.V. Jawahar	ACL Anthology (WSLP Workshop)	Character-level frame boundary annotation (transition-onset rule), ResNet-18/MLP feature extraction from RGB frames or RTMPose hand keypoints	ResNet-BiLSTM / MLP-BiLSTM with CTC + frame-level cross-entropy loss (two-stage training); comparison with ByT5 and Transpeller (BSL-pretrained)	Dense frame supervision: 4.87% CER (standard split), 16.8% CER (signer-independent) - outperforms word-level baseline (16.4%/37.4%) and BSL cross-lingual transfer (8.1%/15.7%); Detection: 84.6% F1
3	2025	Shilpa Ingoley, Jagdish Bakal	Procedia Computer Science (Elsevier)	MediaPipe-based hand landmark (21-point) detection, bounding box cropping without touching hand contour, centering, padding to square, resizing to fixed size	Transfer learning with Vgg16, Vgg19, ResNet50, MobileNetV2 (ImageNet pretrained, frozen base + custom head with batch norm, dropout, dense layers); Adam optimizer, categorical cross-entropy, TensorFlow, Google Colab (T4 GPU)	MobileNetV2: 99.69% validation accuracy (best overall, smallest size 42.8MB); Vgg16: 99.67%; Vgg19: 99.16%; ResNet50: 98.72% (lowest) -- on self-created 35-class ISL dataset (600 images/class)
4	2025	Mayowa O. Oyediran, Kazeem M. Olagunju, Olufemi S. Ojo, Boboye J. Adebayo, Ayodele A. Adebiyi, Emmanuel O. Asani	NIPES - Journal of Science and Technology Research	Grayscale conversion; noise removal (Gaussian technique + median filtering); image resizing; normalization; Canny Edge Detection for segmentation	CNN and CLSTM (Convolutional LSTM) models; hyperparameter optimization via swarm intelligence algorithm, grid search, and Bayesian optimization; MATLAB with custom GUI	CLSTM (optimized): 96.50% accuracy at 41.04 sec recognition time (best); CNN (traditional): 93.57% accuracy at 62.06 sec recognition time -- on 39,000-image, 26-class alphabet dataset
5	2025	Nishtha Bhagyawant, Gauri Tamondkar, Sneha Yadav, Shwethashree Kenche, Sunny Sall	International Journal of Soft Computing and Engineering (IJSCE)	Video frame extraction; MediaPipe/OpenCV-based key point extraction; image resizing; frame selection/difference; background subtraction; frame normalization; data augmentation	CNN-LSTM model (Conv2D, Max Pooling, LSTM, Dense layers) vs SVM-LSTM model; Adam optimizer with learning rate schedule; mixed precision training; early stopping; ISL-Corpus dataset (700 annotated videos, 100 sentences)	CNN-LSTM: 84% training accuracy (outperformed SVM-LSTM), but validation accuracy 100% indicating overfitting (validation loss 5.57); SVM-LSTM: 66% training accuracy (lower)
6	2025	Manish Shukla, Harsh Gupta, Ashish Sharma	Zenodo (Open-source/Technical Report)	Hand histogram generation via HSV color space (H,S channels); histogram back-projection for hand segmentation; grayscale conversion; resizing to 50x50; image flipping (horizontal/vertical) for augmentation; train/val/test split (~70/15/15)	Custom compact CNN (Conv2D layers with 16/32/64 filters, Max-Pooling, Dropout 0.5, Dense+softmax); Keras/TensorFlow, OpenCV, NumPy; SGD optimizer (momentum 0.9), categorical cross-entropy	>95% accuracy on 44 ASL gesture classes (26 letters + space/delete + extras); real-time webcam inference on CPU; no confusion matrix/detailed metrics reported
7	2024	Hafiz Muhammad Sarmad Khan, Irene Murtagh, Simon D. McLoughlin	ELRA (11th Workshop on Representation & Processing of Sign Languages)	Conversion of ISL-HS video frames to Motion History Images (MHIs); data augmentation via Gaussian noise, multiplication, affine transforms (scaling/zooming)	CNNs: ResNet50/101/152-V2, Xception, DenseNet121/169/201, InceptionV3, Inception-ResNetV2 (TensorFlow/Keras, Google Colab, Tesla K80)	DenseNet121 best: 90.38% accuracy; DenseNet201: 90.10%; DenseNet169: 89.60%; lowest was Inception V3 at 75.76%
8	2024	Aziza Zhidebayeva et al.	IJACSA, Vol. 15, No. 9	Frame extraction from parquet/video sequences, keypoint-based feature extraction (facial/hand landmarks), masking, embedding (384-dim)	Two-stream 2D CNN (spatial + temporal), Encoder-Decoder with BiLSTM/Sequential classifier, softmax classification; TensorFlow	DL models consistently outperform traditional ML as training data increases (comparative curve); confusion matrix shows strong diagonal accuracy across 20 classes with some confusion (e.g., classes 10 & 20)

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9	2024	Durgesh M. Sharma, Shahana G. Qureshi, Hemant Kumar Gianey, Kunika Bhutada, Sahil Khune, Vinaykumar Pillai, Kushagra Selokar, Siddhesh Singabhatti, Hitesh Gehani, Shishir Kumar Shandilya, Roshan Bhanuse	Research Square (Preprint)	Skin detection via YCbCr color space (converted from RGB); Gaussian Smoothing for noise removal; Canny Edge Detection for visualization	CNN for feature extraction + SVM (RBF kernel) for classification; self-collected ISL dataset (~900 images, 700 videos, 36,000 greyscale images, 7 signers)	CNN alone: 100% accuracy on test set; SVM classification: >95% accuracy; overall system: 97% accuracy for finger-spelling alphabets, 91% for single-handed dynamic words
10	2023	Zhadra Kozhamkulova et al.	IJACSA, Vol. 14, No. 9	Grayscale conversion, median filtering (noise removal), histogram equalization (lighting normalization), resizing to 512x384, frame-length normalization (25 frames/sequence)	2D CNN (cascaded), Convolution+Activation+Maxpooling blocks, fully connected + dropout layers; Levenshtein distance for evaluation	Histogram of Levenshtein distances skewed toward low error (0-5 range); training/validation loss converge without major overfitting (qualitative results, no single accuracy % reported)
11	2022	Siriwiwat Lata, Sirawan Phiphitphatphaisit, Sarayut Gonwirat, Olarik Surinta	ICIC Express Letters, Part B: Applications, Vol. 13, No. 9	Uniform-distribution frame selection (32 frames/video), YOLOv5-based human/ROI detection	CNN-LSTM architectures: MobileNetV2-LSTM, ResNet50-LSTM, DenseNet201-LSTM (+ GRU comparison, fusion CNN-LSTM); dropout, GAP, softmax; SGD optimizer	ResNet50-LSTM (64 units, no dense layer) best: 88.42% test accuracy, 89.15% (5-CV) on Dynamic Thai Fingerspelling dataset; Fusion (MobileNetV2+ResNet50)-LSTM: 88.62% (marginal, non-significant improvement)
12	2021-22	Brandon Vu, Richard Chen, Nyle Wong	Stanford University (CS231n course project report)	Grayscale, resize to 28x28, OpenCV contour-based & detector-based hand cropping/centering, combined multilingual label space (ASL/JSL/ISL/ArASL)	Faster R-CNN (Detectron2, ResNet-FPN-50) for hand detection; 3-layer CNN + 2 FC layers + dropout for language/letter classification; Adam optimizer	Classifier: 90.25% test accuracy; Hand detector: 94.66% AP(0.5) / 82.39% AP(0.75) on test set; Crop+classify pipeline: 8% accuracy vs 2% without cropping on unseen out-of-distribution dataset
13	2021	Frank Fowley, Anthony Ventresque	CEUR-WS Proceedings (Vol-3105, D-REAL/UCD)	Synthetic RGB & skeletal wireframe image generation via Blender (illumination, viewpoint, size, colour variation); MediaPipe pose estimation applied to test images	VGGNet16, InceptionV3, Xception, ResNet152V2, MobileNetV2 (transfer learning, fine-tuned); Keras/TensorFlow, OpenCV, MediaPipe, Blender	Phase 1 (RGB): 33% accuracy (VGGNet16). Phase 2 (skeletal wireframe + MediaPipe): 71.4% accuracy on ISL-HS corpus; outperformed SOTA in-the-wild (62%)
14	2021	Thongpan Pariwat, Pusadee Seresangtakul	Symmetry (MDPI), Vol. 13, No. 2	Semantic segmentation (SegNet + dilated convolution) for hand segmentation on complex backgrounds; optical flow (Lucas-Kanade) for multi-stroke separation; Gaussian filtering, bounding-box cropping, resize to 150x250	CNN (5 structural variants compared, final based on AlexNet); ReLU, max/avg pooling, FC+softmax classification	Best CNN format (AlexNet-based): 92.03% (5-fold CV); Multi-stroke recognition: One-stroke 88.00%, Two-stroke 85.42%, Three-stroke 75.00%; Overall average 85.60%